

The Second Law of Thermodynamics Ludwig Boltzmann, who spent much of his life studying statistical mechanics. Let's do some combinatorics here. Imagine you have a fair coin, and you flip that coin 3 times. You define an event E .

How many possible outcomes are there inside the sample space Ω (or S)?

The event E is called as the **macrostate** of the system, and each of its elements is called **microstate**. For example, the macrostate "3 heads" has 1 microstate. The event E is called as the **macrostate** of the system, and each of its elements is called **microstate**. For example, the macrostate "3 heads" has 1 microstate. The event E is called as the **macrostate** of the system, and each of its elements is called **microstate**. For example, the macrostate "3 heads" has 1 microstate.

The problem is, N and q are always very large numbers. For example, in 1 gram of N_2 , the number of chemical bonds is up to 10^{23} .

$$\begin{aligned} \ln N! &= \ln [N(N-1) \cdots 2 \cdot 1] = \ln N + \ln(N-1) + \cdots + \ln 2 + \ln 1 \\ &= \sum_{i=1}^N \ln i \approx \int_1^N \ln x dx \approx N \ln N - N \end{aligned}$$

[Stirling's Approximation] When $N \gg 1$:

Recall the definition of Gamma function, for every non-negative integer N , we always have: $\Gamma(N+1) = \int_0^{+\infty} x^N e^{-x} dx$

$$\begin{aligned} &= \int_0^{+\infty} e^{N \ln x} e^{-x} dx = \int_0^{+\infty} e^{N \ln x - x} dx \\ &= \int_0^{+\infty} e^{N \ln x - x} dx \quad \text{Now we want to expand the equation by Taylor expansion, by setting an auxiliary variable } x = y + N \\ &= N \ln N \left(\frac{y}{N} + 1 \right) - (y + N) \\ &= N \ln N + N \left(\frac{y}{N} - \frac{1}{2} \cdot \frac{y^2}{N^2} \right) - y - N \\ &= N \ln N - \frac{1}{2} \cdot \frac{y^2}{N} - N \end{aligned}$$

After changing the variable, we obtain: $\Gamma(N+1) = \int_0^{+\infty} e^{N \ln x - x} dx = \int_{-N}^{+\infty} \exp \left(N \ln N - \frac{1}{2} \cdot \frac{y^2}{N} - N \right) dy$

$$= N^N e^{-N} \int_{-N}^{+\infty} \exp \left(-\frac{1}{2} \cdot \frac{y^2}{N} \right) dy$$

Since N approaches infinity, applying Gaussian integral result:

We conclude: $\Gamma(N+1) = N^N e^{-N} \sqrt{2\pi N} \int_{-\infty}^{+\infty} \exp \left[-\left(\frac{y}{\sqrt{2N}} \right)^2 \right] d \left(\frac{y}{\sqrt{2N}} \right)$

$$= N^N e^{-N} \sqrt{2\pi N}$$

Let's apply our previous results to explore a very cool phenomenon. I have an **interacting system**, where the energy is distributed among N particles. What is the most probable configuration for energy distributing system? We have to find the maximum of $\Omega(N, q)$.

We already assumed that $N_A = 6$, $N_B = 9$, and $q = q_A + q_B = 8$, so:

The multiplicity of an interacting system is: $\Omega_{AB} = \Omega_A \cdot \Omega_B$

As you can see here, the most probable configuration of an interacting system is $q_A = 3$, $q_B = 5$, with the highest multiplicity $\Omega_{AB} = 120$.

Our calculation above seems like make sense; the "equilibrium" state of energy is the most likely reached state. Energy is conserved.

The amount of energy stored is directly proportional to the number of chemical bonds in a solid; to put it more simply, the energy is proportional to the number of particles N .

Take natural logarithm of both sides, and use Theorem 3.1: $\ln \Omega(N, q) = \ln(q+N)! - \ln q! - \ln N! = (q+N) \ln(q+N) - (q+N) \ln(q+N) - q \ln q - N \ln N$. Taking natural logarithm of the multiplicity of an interacting system yields: $\ln \Omega_{AB} = \ln \Omega_A + \ln \Omega_B$.